

Does DEIA Still Matter?: A Survey of VR Researchers at IEEE VR

Aleshia Hayes*

University of North Texas

John Quarles†

University of Texas San Antonio

Greg Welch‡

University of Central Florida

Dylan Fox††

XR Access

ABSTRACT

This study explores perspectives on diversity, equity, inclusion, and accessibility (DEIA) from researchers in the IEEE VR community. Fourteen participants expressed sustained commitment to DEIA, noting underrepresentation of women, BIPOC, individuals with disabilities, and researchers from developing countries or low socioeconomic backgrounds. Participants perceived high costs, lack of accessibility features (e.g., subtitles, mobility support), and insufficient family resources (e.g., childcare) as barriers. Among factors that could deter support of DEIA listed were fear of retaliation, financial constraints, structural challenges, and difficulties identifying diverse representatives. Strategies participants supported included reporting demographic statistics, strategic planning, and promoting DEIA in speakers and attendees.

Index terms: Diversity, Equity, Inclusion, Accessibility, Broadening Participation.

1 INTRODUCTION

In recent history, diversity, equity, inclusion and accessibility efforts have been supported by organizations and companies as profitable and beneficial to workforce climates. Despite years of research that has demonstrated the positive impact of diverse teams on research outcomes and profit, the global climate towards diversity, equity, inclusion, and accessibility (DEIA) has moved from a pragmatic consideration to a political talking point. By 2024, Florida, Texas, and Utah had banned diversity, equity, and inclusion efforts in higher education and publicly funded organizations. Aligning with President Trump's 2020 "Executive Order on Combating Race and Sex Stereotyping". The following year, President Biden overrode the Trump order with his "Executive Order on Diversity, Equity, Inclusion, and Accessibility" in the Federal Workforce. Notably, the Trump 2020 Executive Order did not mention accessibility, while the Biden Executive Order refers to Diversity, Equity, Inclusion, and Accessibility (DEIA). The January 20, 2025 executive order by President Trump "Ending Illegal Discrimination And Restoring Merit-Based Opportunity" included accessibility or DEIA [5].

The exploratory research in this discourse explores the perspectives on DEIA reported by VR researchers in the research community. To this end, we surveyed researchers who have organized IEEE VR, a major international technology conference.

2 RELATED WORKS

The US National Center for Science and Engineering Statistics (NCSES) have identified the disparities in the population of the United States and representation in STEM workforce. For instance, they report that 51% of the US population is female, only 38% of

the STEM workforce is female [1]. Similarly, people with one disability make up 9% of the US population, but only 3% of the STEM workforce.

Research has highlighted one cost of the lack of diversity among development and research teams as the exclusion of underrepresented populations and groups from engagement in the growth of the new emerging technologies. Peck et al. [2] identified a trend of virtual reality technologies catering to a certain demographic they label M-WEIRD, male, white, educated, industrialized, rich, and democratic populations - M-WEIRD [2].

Fox and Thorton [3] discuss concerns that extended reality technologies have been catered to able-bodied individuals and neglect the needs of those with disabilities. In addition to the inaccessibility of extended reality for those with disabilities, systematic issues such as poverty, racial discrimination, and digital divide, and limited technology access also cause more unequal user experiences [3]. This exploratory study looks at the perspectives about DEIA reported by leaders and organizers of the IEEE VR international conference, over the past decade. This paper outlines practices reported by conference general chairs, doctoral consortium chairs, and diversity chairs that have addressed diversity, equity, inclusion, and accessibility in the IEEE VR conference in the past decade. This question is timely, as many companies are shifting their messaging or intent to support DEIA initiatives, IEEE VR has remained committed to DEIA.

The current trend of ending DEIA programming in the face of technological giants like Facebook supporting the end of DEI while Apple is maintaining its commitment to diversity leads us to ask the question, "is DEI still important to the IEEE VR Community?"

3 METHODS

They were then asked to rate similar efforts taken by themselves and their employers. Next, participants were asked if they have personally taken steps to increase DEIA in their general professional roles.

Similarly, participants were asked open ended questions about who they believe is being excluded (if anyone), what they wished the community would do differently, and the quality and effectiveness of the XR community's efforts to support and advance DEIA. Finally, participants were asked to identify reasons that may deter themselves or others from supporting DEIA efforts in general or at professional conferences.

Twenty-five people began the survey, and fourteen people completed the survey. Given the population of approximately 150 leaders of the IEEEVR conference in the past decade, the sample size is sufficient for a 90% confidence level with a 20% margin of error. Of the 14 participants who completed the survey, 11 were male and three were women. 47% had served as a general chair, 53% had served as a program chair, 33% had served as Doctoral Consortium or Future Faculty, 27% had served as DEIA chairs, and 27% served as chairs of publications, posters, or workshops.

3.1 Apparatus

The survey for this study asked questions that could inform the practice of Diversity, Equity, Inclusion, and Accessibility from the perspective of the career researchers and scientists who engage with the IEEE VR community. In addition to demographics questions, there were questions about the current practices related to DEIA.

* e-mail: Aleshia.prof@gmail.com

† e-mail: john.quarles@utsa.edu

‡ e-mail: welch@ucf.edu

†† e-mail: dylan@xraccess.org

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Finally, participants were asked their rating of usefulness and viability of the 2021 United States Office of Personnel Management (OPM) created a strategic plan focused on enhancing diversity, equity, and inclusion. The plan includes several key areas of action, such as improving demographic data collection of employees in order to improve the power of data to inform decisions.

Other items included deeper questions of personal perspectives regarding DEIA, "If you identify a lack of diversity, equity, and inclusion in the community, what groups do you believe are not being equitably represented in our community?" Participants were also asked, "Are there any reasons that may deter you or others from supporting DEIA efforts in general or at professional conferences?"

4 RESULTS

In the survey, thirteen participants answered the question, "Have you, personally, taken steps to increase the DEIA of the IEEE VR conferences over the past 10 years?" Three respondents reported that they have taken many steps, eight participants indicated they have taken "some" steps, and two participants reported that they have taken no steps to improve DEIA of the IEEE VR conference. The question "How would you characterize our community's efforts to improve DEIA in the past decade?" All twelve participants who answered this question chose the neutral option on a Likert scale, indicating "The community has taken 'some' steps." None of the participants indicated that the community has taken "many" steps, but none of the participants indicated the community has taken "no" steps. Similarly, they indicated that the conference's approach to diversity has been intermittently.

Eleven of the fourteen participants answered the open-ended question, "If you identify a lack of diversity, equity, and inclusion in the community, what groups do you believe are not being equitably represented in our community?" The synthesis of responses includes underrepresentation of women, BIPOC, people with disabilities, and researchers from developing countries or low socioeconomic backgrounds. Respondents also identified high costs, lack of accessibility features (e.g., subtitles, mobility support), and insufficient resources for families (e.g., childcare) as barriers for these groups.

Finally, ten participants responded to "Are there any reasons that may deter you or others from supporting DEIA efforts in general or at professional conferences?" Four participants did not answer, and two participants indicated that there were no reasons. The other eight participants indicated that fear of retaliation, financial constraints (e.g., childcare, accessibility needs), structural challenges, such as limited funding, the small size of some communities, and difficulties in identifying qualified representatives for diverse groups, and a lack of knowledge about how to be the most effective factors that deter them or others from supporting DEIA professionally.

5 CONCLUSION

The most surprising findings were the open-ended responses. Of note was the reported fear of retaliation for being involved in DEI programming. Given the public separation of DEI and accessibility, it is notable that the Executive Order stated that the "policy of the United States not to promote race or sex stereotyping or scapegoating in the Federal workforce or in the Uniformed Services, and not to allow grant funds to be used for these purposes". It should be noted that the state bans on DEI programming are not banning accessibility efforts. Similarly, the Americans with Disabilities Act has not been challenged. It will be up to the leadership of the IEEE VR and other technology conferences to decide how they will move forward to support the

profitability and effectiveness produced by diverse research and development teams. Overall, the 2021 standards were rated favorably in terms of being worth the effort (return on investment) and in terms of being feasible.

6 DISCUSSION

This study outlines perspectives reported by conference general chairs, doctoral consortium chairs, and diversity chairs that have addressed diversity, equity, inclusion, and accessibility in the IEEE VR conference in recent years. Given annual changes in conference leadership, values and practices are not always passed on from year to year. This makes the responses that the participants mostly indicated the IEEE VR conference has been intermittent. The current study is, of course, limited by the small sample size and the narrow focus of the researchers in the IEEE VR community.

This paper will be presented at the 2025 IEEE VR conference, following the 2025 executive order by US President Trump. "Ending Illegal Discrimination And Restoring Merit-Based Opportunity" that calls for the end of "diversity, equity, inclusion, and accessibility" (DEIA) that can violate the civil-rights laws [5, p. 1]. This executive order validates the fact that participants of this study listed fear of retaliation as a deterrent from supporting DEIA. While science has demonstrated the efficacy and profitability of DEIA, public and political discourse suggests a gap in sharing this knowledge. The 2025 executive order also calls for the end of DEI principles and policies in the private sectors, enforced by lawsuits.

Future research should explore this topic more broadly among STEM professionals and fields. The participant pool was skewed male, as is consistent with the rates of representation in communities of computer science researchers and developers, which exemplifies the need for diverse participants. Likewise, future research should expand to include deeper questions of the "highly effective practices" that leaders in STEM fields have designed or experienced.

Given the value of diversity acknowledged by the IEEE VR community and leadership, documentation and consistency of the community efforts that have been established (diversity committees, mentoring programs, funding, active recruitment from diverse pools). Data collection and analysis could also move the community towards more evidence-based practice that can inform community, conference, and organizational levels of leadership on the costs and benefits of DEIA programming for the community.

A KNOWLEDGEMENTS

We'd like to thank the members of the SURGE XR lab and special thanks to Kathleen Dang who supported the literature review.

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